CSCI160 Computer Science I
Lab Exercise: Days of the Week

Objective: In this lab exercise, you will:

1. Use VS Code and the terminal to write, execute and test a Python script
2. Practice using data structures to organize memory for efficient use
3. Use a provided Python module to call a function and process the results
4. Continue exploring problems with modulus division
5. Define test cases for your algorithms

AI Usage: In order to get the most out of this exercise you should try to solve it yourself without using AI. The only way to strengthen coding and problem solving skills is through active struggle. If you are stuck, resist the temptation to simply ask for the solution. Engage with AI with very specific single task oriented prompts and ensure that you fully understand all suggestions, and verify that they are correct.

Documentation: Any AI code used in this lab ***must be cited.*** Code comments are required! If you wrote something yourself, document that. If you used AI, document that as well. Remember, your submission of this lab is a statement that you wrote every single character that does not have an AI citation and that you 100% understand every line.

While working on this lab you will be asked to

- Take screen shots. Please put them in a single document. You will be submitting the document when finished, along with your Python script file
- Download a custom Python module from Brightspace called: `zellers.py` you will be importing this module to use the function defined in the file. Make sure you put this Python file in the same folder as your work. You are not expected to understand the code here, just to be able to use it properly. Refer to Week 3 in zyBooks for proper module importing and function usage. **Do Not Copy The Code Into Your Own Script!!**

Problem Overview:

You want to write a Python program that will allow users to enter a western style date in `MM/DD/YYYY` format. This date will mark the departure day of a vacation. The program should then display the day of the week.

You have been provided with the **Zeller's Congruence** algorithm to determine the day of the week for a given date.

After that, you want to enter the number of days that you'll be gone on vacation. The program will then tell you the following

- The day of the week that you will return
- The number of weeks and days you will be gone

Here is an example

```
Enter the date you are leaving for vacation (MM/DD/YYYY): 02/16/2025
You are leaving on a Sunday.

Enter the number of days you'll be gone: 23
You will return on Tuesday
You will be gone for 3 weeks and 2 days.
```

Assumption: The day you leave is not counted

Understand that the program should create correct results for *any input*

Part 1 – Memory Organization with data structures: Day of the Week Mapping

1. **Assumption:** Saturday is the first day of the week. We are using Saturday as the first day because we will be using a mathematical formula that treats the days of the week in this fashion
2. **Concept:** Programmers often like to order items in a sequence using `zero-based positions: Saturday = 0, Sunday = 1 . . . Friday = 6`
3. We want to organize our data so that when given the number (int) of the day of a week, we can easily determine the name of the day (string).
4. We can conveniently associate the days of the week with numbers using a `list.` This approach uses a single variable to organize multiple values into memory.

```
days = ["Saturday","Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday"]
```

0	1	2	3	4	5	6
"Saturday"	"Sunday"	"Monday"	"Tuesday"	"Wednesday"	"Thursday"	"Friday"

5. It will also be convenient to organize our data so that we can convert the name of the day (string) to it's number (int).
6. We'll use a `dictionary` for this

```
day_numbers = {
    "Saturday" : 0,    # Saturday is the first day of the week
    "Sunday"   : 1,    # Sunday   is the second day of the week
    "Monday"   : 2,    # Monday   is the third day of the week
    "Tuesday"  : 3,    # Tuesday  is the fourth day of the week
    "Wednesday": 4,    # Wednesday is the fifth day of the week
    "Thursday" : 5,    # Thursday is the sixth day of the week
    "Friday"   : 6,    # Friday   is the seventh day of the week
}
```

Task One: Create a new Python file and name it: `vacation.py`

1. Add your name header to the file (name, date, lab problem description)
2. Define the data structures shown above
3. **Experiment with**
    1. Using a number to select the name of a day of the week. ***Print the results***
    2. Using the name of the day of the week to select its number. ***Print the results***
    3. For those of you who know if statements: **`DO NOT USE THEM!!`**
    4. **Milestone #1:** Screenshot the results and paste the screenshot into your document, similar to the following. Also show the code in the screen shot.

```
Enter the day of the week as a number: 4
The day of the week is: Wednesday

Enter the day of the week as a string: Wednesday
The day of the week is: 4
```

    5. If you are unsure how to accomplish this you should do the following
        1. Revisit the zyBooks lesson on **Accessing list elements**
        2. Revisit the zyBooks lesson on **Accessing dictionary entries**
        3. Ask AI to provide you with examples and explanations. ***If you choose to do this screen shot your conversations and paste into your submission document.***

Part 2 – Using a custom module: Zeller's Congruence for Day of the Week Calculation

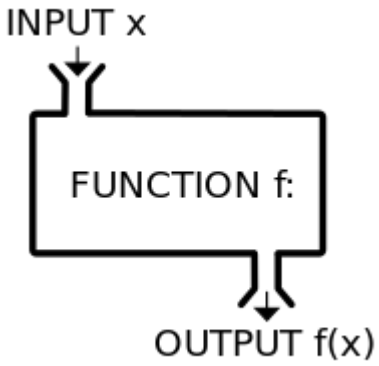
You have been provided with a custom module in the file `zellers.py` that contains the `zellers_congruence()` function. This function will help you determine the numeric day of the week for any given date in the Gregorian calendar.

Example: convert the integers: `day, month and year  ==> integer day of the week`

- `zellers(16, 2, 2025) ==> 1 (Sunday)`

Task One: Download and use the `zellers.py` module

1. Download the file and place it in the directory with your Python script
2. **Note:** You are not expected to understand or even attempt to understand this algorithm (unless you want to). Module functions can be treated as `black boxes.` You only need to know how to use the function. This involves
    1. Calling the function properly using its name
    2. Providing correct function input
    3. Handling the function output



3. Examining the `zellers_congruence` we can see that it requires the integers: day, month and year as input and it will return the day of the week as an integer

```
def zellers_congruence(day: int, month: int, year:int) -> int:
    '''
    Calculate the day of the week using Zeller's Congruence.

    Parameters:
        day     (int): The day of the month.
        month   (int): The month of the year.
        year    (int): The year.
    Returns:
        int: The day of the week as an integer.
    '''
```

4. **Milestone #2:** In your `vacation.py` script import the `zellers` module and call the `zellers_congruence` function with some test cases of `day, month and year` and print the output as an integer and as the string day of the week.

Here is an example

```
Using 15, 2 and 2025 the day of the week is: Saturday
Using 15, 2 and 2025 the day of the week is: 0
```

Screenshot the results and paste the screenshot into your document, similar to the following. Also show the code in the screen shot.

Part 3 – User input: Handling the user input of the date

Goal: Allow the user to enter a full date in `"MM/DD/YYYY"` format, but in order to use the `zellers_congruence` the program will need to ***slice the date string into the integers*** month, day and year

Here is an example

```
Enter the date you are leaving for vacation (MM/DD/YYYY): 02/15/2025

The date 02/15/2025 is 2, 15, 2025.
```

1. **Assumption:** The date will always have 2 digits for the day and month and 4 for the year
2. **Implementation:** Use `string slicing` to accomplish this. Remember that the individual characters are indexed starting with 0.

   Review the zyBooks section **4.3** on **String slicing**

   Use the following table to help understand where the slices should begin and end.

Date String	0	2	/	1	5	/	2	0	2	5
Index	0	1	2	3	4	5	6	7	8	9

Part 4 – Calculating the day of return: Given the day of departure (integer) and the number of days gone, what day of the week will you return?

1. This is another use of modulus division. To calculate the result, take the day of the week of the departure (integer) and add it to the number of vacation days.
   1. Using the example above:
      1. You left on a Sunday, which is day 1.
      2. You were gone for 23 days. `23 + 1 = 24`
      3. How many weeks is in 24 days: `3 weeks`
      4. How many days are left over: `24 % 7 = 3`
      5. The third day of the week is `Tuesday`
2. Implement this algorithm and display
   1. The name of the day of the week that you'll return, using the list to extract the correct string
   2. How many weeks and how many days you'll be gone. Refer to the example above for formatting.

Part 5 – Test Cases:

Here are 3 test cases. Use these to test your algorithm.

Input	Output
02/16/2025 23	Leave: Sunday Return: Tuesday 3 weeks and 2 days
01/01/2025 7	Leave: Wednesday Return: Wednesday 1 week and 0 days
12/30/2025 1	Leave: Tuesday Return: Wednesday 0 weeks and 1 day

Task: Define 5 test cases of your own. Write these as comments in your code.

Submission: Attach your Python file and your submission document containing the screen shots.

Review the rubric to ensure you met all requirements.